Turning Japanese financial-statement PDFs into correct tables

A pipeline design for robust, verifiable extraction, with evidence from a working prototype

Take-home submission, 27 October 2026

The problem. Japanese listed companies publish their financial statements as PDFs: earnings releases (決算短信), annual securities reports (有価証券報告書), half-year reports and corrections. The pages vary in layout, era, accounting standard, industry and production quality, from clean digital output to scans. The goal is to convert each statement page into a table that is *correct*, and to define and measure what correct means.

The central idea. A financial table is not correct because it looks right. It is correct when every number is grounded in evidence on the page, consistent with the accounting relationships it must satisfy, and in agreement with independent sources wherever they exist. The pipeline is built around those three tests, and it abstains rather than guesses when they cannot be met.

Key decisions. The unit of correctness is the *fact*: row identity, period, context, unit and typed value. Two extraction paths with independent failure modes (geometry over the PDF's text layer, vision over rendered pixels) feed one typed intermediate representation. Verification constraints are derived from each table's own hierarchy and from the EDINET taxonomy's calculation relationships, and checked with interval arithmetic so that rounding never raises false alarms. Ground truth at scale comes from the filings' own XBRL. Every page ends in one of four explicit states.

74 / 74	20 / 20	13 of 20	45%
facts correct on the sample page (prototype)	derived accounting constraints pass, 100% cell coverage	false alarms a naive equality check raises on the same correct page	of adjacent-row value swaps caught by arithmetic alone: the case for redundancy

1. Scope and what this document covers

The input is any page from a Japanese corporate filing that contains a primary financial statement: the balance sheet (貸借対照表, or 財政状態計算書 under IFRS), income statement and comprehensive income statement, statement of changes in equity, and cash-flow statement, in consolidated and non-consolidated form. The output is a structured table per statement, plus a set of typed facts with provenance. Notes tables (segment information, fixed-asset schedules) share most of the machinery and are treated as a later phase.

The sample page provided (page 4 of a corrected earnings release, the asset side of a consolidated balance sheet) is a useful but easy case: digital, single table, two periods, clean indentation. Section 3 catalogues what the wider corpus contains. Sections 4 to 7 describe the design, and Section 8 reports what the prototype demonstrates and where its measured limits are.

2. What “correct” means, and how to measure it

Correctness has three layers. **Content:** every printed token in the statement region is captured exactly. **Structure:** the grid, the row hierarchy, and how each cell binds to its column header and unit. **Facts:** the meaning that downstream users consume. The third layer is the one to measure hardest, because a table can be perfect in the

first two and still unusable.

> **Definition.** A statement page is correct when (i) every monetary cell printed in the statement region appears exactly once as a fact; (ii) each fact carries the right row identity (label and hierarchy path, and the taxonomy concept where one exists), the right period (end date, instant or duration, consolidated or non-consolidated, before or after correction), the right unit and scale, and the right value including sign and the distinction between nil, blank and zero; and (iii) no fact exists that is not printed. Page metadata (statement type, unit line, accounting standard) is part of correctness.

Three distinctions in that definition matter in practice. A dash (－) means the line exists with no balance; a blank means the line is not presented for that period; they are different facts. A printed △0 is a negative amount smaller than one display unit, so its sign must survive. And identity must be the path, not the label: the sample page alone has four rows labelled その他 and four labelled 減価償却累計額.

Metrics

Metric	Definition	Role
Fact precision, recall, F1	Exact match on (row path or concept, period, context, typed value, unit)	Headline accuracy
Page-exact rate	Share of pages with zero fact errors and correct metadata	The strict number users experience: one wrong digit makes a statement wrong
Risk and coverage	Error rate among auto-accepted facts, against the share of facts auto-accepted	The operating point; both numbers are reported together
GriTS and TEDS	Grid similarity of predicted and true tables [1][2]	Diagnostics, and comparability with published table-recognition results
Hierarchy accuracy	Share of rows with the correct parent	Diagnostic for indentation and role errors
Constraint pass rate and coverage	Share of derived accounting constraints satisfied, and share of cells covered by at least one	Ground-truth-free signal for production, valid only alongside its measured detection power (Section 6)

Structure metrics are necessary but not sufficient. TEDS compares tables as HTML trees and was designed to penalise cell misalignment and recognition errors [1]; GriTS compares them as matrices and covers topology, location and content together [2]. Neither knows that a column is a period or that a row is a subtotal. On the sample page, pdfplumber's default table extraction reproduces all 43 body rows with their values, so it would score near-perfectly on cell content, yet it drops both period headers, the unit and the entire hierarchy. Its table cannot answer “what was cash at 31 March 2026, in yen?”.

Statistical confidence. If a test set of n independent pages contains no errors, the one-sided 95% upper bound on the page error rate is roughly $3/n$ (the rule of three). Claiming a page-exact rate of 99% with 95% confidence therefore needs about 300 error-free pages *per stratum* that matters, which drives the evaluation design in Section 7. Proposed launch targets, to be calibrated on real data: fact error rate among auto-accepted facts at or below 0.1%, with at least 90% of digital J-GAAP statement pages fully auto-accepted.

3. Where it breaks: a catalogue of complexity

The design starts from the failure modes, not from a tool. The catalogue below was assembled from the sample page, the filing specifications, published research and known properties of Japanese PDF production. Each row names what breaks if the issue is ignored and which stage of the pipeline (Section 5) handles it.

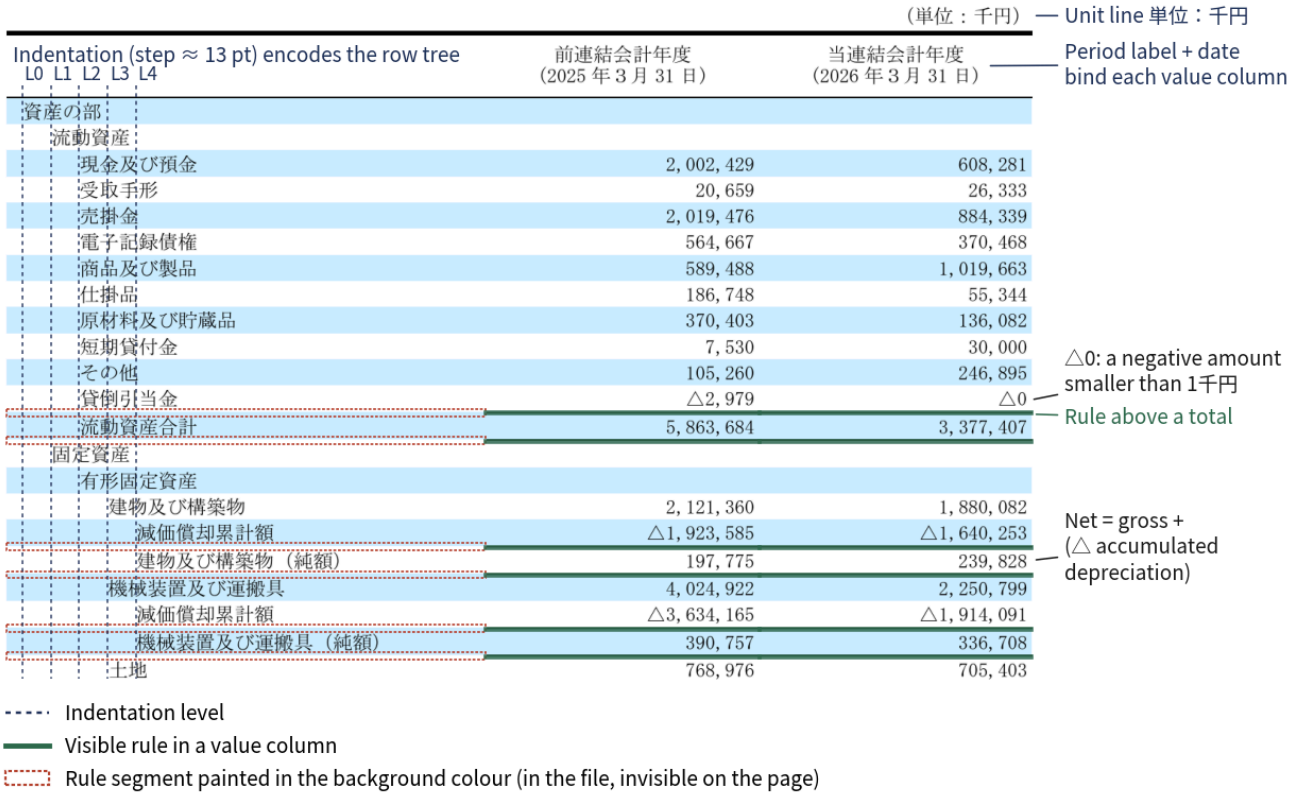


Figure 1. Cues and traps on the sample page. Indentation encodes the row tree; a visible rule sits above every total; period labels and dates bind the two value columns. Rule segments over the label column are drawn in the band or white colour, so a parser that ignores colour sees lines no reader sees. Two current-year cells (投資有価証券, 関係会社株式, not shown in this crop) carry short underlines, the convention for corrected figures in a 訂正後 filing.

Complexity	Example	What breaks if ignored	Handling (stage)
Filing and document			
Filing-type mix	決算短信, 有価証券報告書, 半期報告書, 計算書類, 訂正 filings	Different layouts, comparatives and page furniture	Triage records filing type; per-type layout priors (S1)
Consolidated and non-consolidated	連結 and 個別 statements in one filing with near-identical labels	Facts attached to the wrong entity scope	Title parsing; context flag on every fact (S1, S5)
Accounting standard	J-GAAP, IFRS (連結財政状態計算書, 注記 column), US GAAP. IFRS companies, adopted or decided, numbered 318 and held 51.2% of TSE market capitalisation in mid-2026 [18]	Statement names, label sets and arithmetic all differ	Standard detection; separate concept sets and constraint templates (S1, S5, S6)
Industry formats	Banks, insurers, securities firms, construction, railways; the EDINET taxonomy carries industry-specific prefixes [14]	Hard-coded account lists and rules fail	Industry-aware concept mapping; constraints from taxonomy, not code (S5, S6)

Complexity	Example	What breaks if ignored	Handling (stage)
Corrections	訂正前 and 訂正後 versions; changed cells underlined	Stale values; correction marks misread as summation rules	Corrected-cell detection; diff check against the original (S3, S6)
Restatements and reclassifications	遡及修正, 組替え: the prior-year column differs from last year's published figure	Cross-period checks raise false failures	Cross-period check is soft; mismatches are explained, not failed (S6)
File and text layer			
Raster content	Scanned filings; images pasted into otherwise digital PDFs	No glyphs to read	Region-level glyph check routes to vision (S1, S3b)
Invisible OCR layers	Scans carrying a hidden text layer of unknown quality	Confidently wrong text treated as ground truth	Detect image-under-invisible-text; treat the layer as a hint only (S1)
Broken Unicode maps	CID fonts without, or with wrong, ToUnicode tables	Garbage text, or plausible but wrong digits	Text-layer audit against pixels (S1)
Look-alike code points	Kangxi radical characters (U+2Fxx) in place of unified ideographs, which looks identical on screen [19]	Label matching silently fails	NFKC normalisation; audit flags non-standard blocks (S3a)
Simulated bold	Glyphs drawn twice with a small offset [20]	Doubled characters such as 流流動動資資産産	Character de-duplication before word building (S3a)
Invisible graphics and text	Background-coloured rules (Figure 1), white text, clipped text	Phantom lines and cells	Colour-, clip- and render-mode-aware geometry (S3a)
Fragmented or misordered runs	Content order differs from visual order; numbers split into runs such as “2,” “002,” “429”	Merged or split cells	Geometric ordering; numeric fragment merging (S3a)
Rotation	Landscape pages for wide statements (株主資本等変動計算書)	Rows read as columns	Orientation detection and de-rotation (S1)
Page layout			
Account form (勘定式)	Assets on the left, liabilities and net assets on the right of one page	Two tables interleaved row by row	Detect two label columns; split into two logical tables (S2)
Several tables per page	Income statement and comprehensive income statement together; statement plus notes	Tables merged into one	Segmentation by titles, unit lines and rules (S2)
Multi-page statements	Balance sheet over two pages; header repeated or not; （続き）; unit stated only on page one	Truncated statements; orphan pages without unit or periods	Cross-page assembly with inheritance [4][5][8] (S4)
Page furniture	Running headers, page numbers, （訂正後） marks	Spurious rows	Positional and pattern filters (S2)
Table structure			
Spanning column headers	Equity statement: 株主資本 over 資本金, 資本剰余金, 利益剰余金, 自己株式	Columns lose their meaning	Header tree; every column gets a path (S3)
Non-monetary columns	注記番号 (※1), 増減 change columns, 構成比 percentages	References and ratios parsed as amounts	Column typing: monetary, ratio, reference (S3, S5)

Complexity	Example	What breaks if ignored	Handling (stage)
Section rows without values	資産の部, 流動資産: “projected row headers”, which FinTabNet annotations nearly always over-segment [3]	Headers merged into data rows, or treated as data	Explicit section role; never merged with neighbours (S3, S5)
Wrapped and spaced labels	Long labels over two lines; 均等割付 spacing such as 現　金　及び　預　金	One row becomes two; labels fail to match	Continuation-line merging; intra-label spaces removed after NFKC (S3a)
Leaders and decoration	Dot leaders (……) in older filings; bold totals	Noise inside labels; missed totals	Leader stripping; bold as a role cue (S3)
Weak or no indentation	Flat lists with bold subtotals; inconsistent indents	Wrong hierarchy, so wrong constraints	Combine indentation, lexical cues (合計, 小計, 純額) and the taxonomy presentation tree (S5)
Cells and numbers			
Negative notations	△, ▲, (1,234), −1,234; △ confused with Greek Δ by OCR	Sign errors	Normalisation table; OCR confusion set (S5)
Nil, blank, zero	ー ― — ﹣, empty cell, 0, △0	Distinct facts collapsed	Typed nil and blank; sign kept for △0 (S5)
Units and scale	円, 千円, 百万円; per-share amounts in 円 with decimals inside a 百万円 statement; percentages	Values off by 1,000; a scale error is invisible to sum checks	Per-row unit override; cross-document scale checks (S5, S6)
Rounding mode	Truncation (切り捨て) or rounding (四捨五入) to the display unit	Totals never equal their sums exactly (13 of 20 on the sample)	Interval arithmetic; rounding mode inferred per document (S6)
Markers attached to values	※1, *2 next to amounts or labels; correction underlines	Digits absorbed into amounts	Marker tokens separated and kept as references (S3)
Semantics			
Same label, different meaning	その他 four times on one page	Facts overwrite each other	Identity is path plus concept, never label alone (S5)
Label variants and company items	Revenue-standard wording changes; company-specific extension items	Mapping errors	Standard labels first, then fuzzy and model-ranked candidates; extensions flagged (S5)
Non-tree arithmetic	Income-statement step-downs (売上総利益 = 売上高 − 売上原価); cash-flow 小計; equity-statement matrix	Tree-sum checks do not apply	Constraints from taxonomy calculation links and statement templates (S6)
Periods	和暦 dates (令和7年3月31日); irregular fiscal years; cumulative and three-month quarter columns; instant and duration	Values attached to the wrong period	Period parser; instant and duration typing (S5)

4. Design principles

Seven principles follow from the catalogue. They are what make the pipeline sound rather than merely accurate on the cases it has seen.

Evidence before inference. Every digit in the output must trace to evidence on the page: text-layer glyphs, or OCR tokens with bounding boxes. Models may propose structure (which cell belongs to which row and column) but never author a number. A grounding check rejects any value that cannot be matched to evidence inside its cell's box. Language and vision models are useful here, and numeric hallucinations are exactly the errors that look plausible; published comparisons of multimodal models and classical pipelines also find missing and extra rows to be a principal failure mode [22].

Independent redundancy. Two extraction paths whose errors are unlikely to coincide: a geometry path that reads the PDF's text layer and vector graphics, and a vision path that reads rendered pixels. An encoding fault in the text layer does not affect the pixels, and a recognition fault on the pixels does not affect the text layer. Agreement is strong evidence and disagreement is localised to specific cells. This is the logic of recogniser output voting, long established in speech and handwriting recognition [21].

Verify with the domain, and measure the verifier. Accounting identities are free test oracles: totals, net amounts and cross-statement ties must hold. Their power is not assumed; it is measured by injecting realistic errors and counting what the checks catch (Section 8). That tells us precisely what “all checks pass” guarantees and what it does not.

Abstain explicitly. Every page ends in one of four states: verified, accepted with warnings, needs review, or failed. Downstream systems never receive an unverified number without its flag.

One typed intermediate representation. All stages read and write the same representation (tokens, rows, cells, facts), each element carrying its bounding box and source. Stages can be tested in isolation, either path can be swapped for a better model, and the review interface can show exactly where a value came from.

Provenance and reproducibility. Content hashes for inputs, versioned models, taxonomy and configuration, deterministic settings, and a provenance record on every fact. Any output can be regenerated and explained.

Measure per stratum, not on average. Failures cluster by layout family, era, standard and scan quality. An average over the corpus hides a family that fails completely; the evaluation and monitoring plans are stratified for that reason.

5. Architecture

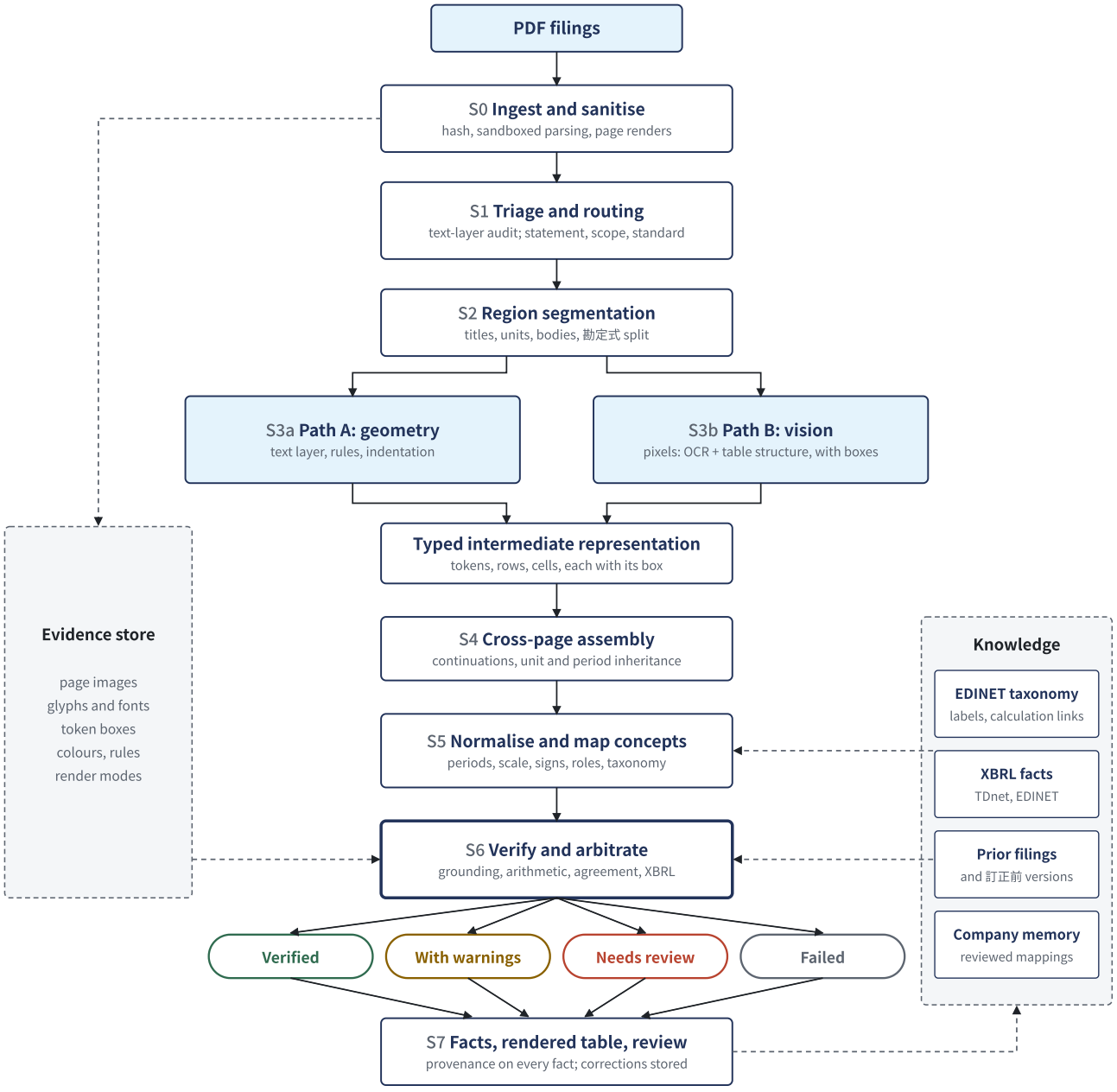


Figure 2. Pipeline overview. Both extraction paths emit the same typed representation. Verification draws on the evidence store (page images, glyphs, boxes) and on external knowledge: the EDINET taxonomy, the filing's XBRL where available, prior filings, and mappings confirmed in earlier reviews.

S0 Ingest and sanitise

Hash every file and de-duplicate. Parse inside a sandbox with time and memory limits, because PDFs are untrusted input. Record producer metadata and page geometry; render pages lazily at two resolutions (fast triage, high-resolution recognition).

S1 Triage and routing

Score each page's text layer before trusting it: share of characters in expected Unicode blocks, replacement and private-use characters, Kangxi radicals, duplicate-glyph rate, and invisible text drawn over images. Then audit it: run a digits-only recogniser over the value-column crops and compare with the text-layer digits. A text layer can render correctly and still extract wrongly, so this cheap check catches broken maps that look fine. Classify the statement from its title (貸借対照表, 財政状態計算書, 損益計算書, 包括利益計算書, 株主資本等変動計算書, キャッシュ・フロー計算書 and their IFRS counterparts) with a layout classifier as fallback, and detect scope, standard, filing type, correction status and orientation. Route the page: geometry-primary when the audited text layer is sound, vision-primary when it is absent or untrusted, both when stakes or uncertainty are high.

S2 Region segmentation

Locate titles, unit lines, table bodies, notes and page furniture. Split account-form balance sheets into their two logical tables and separate multiple statements on one page. On digital pages this uses titles, rule geometry and whitespace; on raster pages, a layout model such as the DocLayNet-trained detector in Docling [6].

S3a Path A: geometry over the text layer

Work at character level: de-duplicate simulated bold, normalise with NFKC, and drop invisible content using fill colour, clipping and text render mode. Build lines by baseline clustering, merge spaced (均等割付) labels and wrapped continuation lines, and discover value columns by clustering right edges of numeric tokens and aligning them with header labels. Build a header tree for spanning headers, quantise indentation into levels, and extract visual cues: rules above totals, double rules, correction underlines (glyph-width, directly under a figure), bold weight. This is the path the prototype implements.

S3b Path B: vision over pixels

Render at 200 to 300 dpi and run recognition with table-structure output. Candidates include PaddleOCR-VL, a compact document model whose latest release reports state-of-the-art results on OmniDocBench v1.6 [7]; MinerU [8]; Docling's TableFormer [6]; and YomiToku, which trains detection, recognition, layout and table-structure models on Japanese documents but is licensed for non-commercial use only [9]. Two requirements apply to whichever is chosen: the output must include token boxes (for grounding and indentation), and any model output without boxes is re-grounded by matching its numbers to recognised tokens. Published scores use TEDS, so they say nothing direct about fact accuracy on Japanese statements; the choice is made on our own stratified evaluation.

S4 Cross-page assembly

Decide whether a table continues from the previous page using several signals: the same statement title or （続き）, groups left unclosed at the bottom of the previous page, compatible columns, and a missing unit line. Inherit unit, periods and column structure. Multi-page tables are now an active research topic: a recent benchmark reports that adding continuation prediction to a page-level extractor cut content error by 47% [4][5], and current parsers merge fragments using column-compatibility heuristics [8].

S5 Normalisation and semantics

Parse periods (Gregorian and 和暦 dates, instant or duration, scope, correction status). Convert values to typed amounts with their uncertainty interval, sign and nil or blank state, and apply unit scale with per-row overrides. Assign row roles (section, item, gross, contra, net, total). Map rows to taxonomy concepts: the EDINET taxonomy provides Japanese and English labels for every element and encodes display hierarchy and calculation relationships [14], and the FSA kept a rule that standard labels may not be overridden for statement items [15], so standard rows can usually be matched exactly. Duplicated labels are resolved by path, company-specific items by ranked candidates, with confirmed mappings stored per company.

S6 Verification and arbitration

Apply grounding, structural invariants, arithmetic constraints, path agreement, cross-document checks and XBRL comparison, then assign the page state. Section 6 details each.

S7 Output, review and learning

Emit canonical facts and a rendered table (schema in Appendix A). Pages that need review open in an interface showing the page crop, both paths' readings and the failing check. Every correction is stored as a labelled example and, where relevant, as a company-level memory (label mappings, layout family) that stabilises the next filing from the same company.

6. Verification: turning accounting into tests

6.1 Interval arithmetic instead of tolerances

Japanese statements show amounts truncated or rounded to the display unit, so totals rarely equal the sum of their printed parts. On the sample page, 流動資産合計 exceeds the sum of its ten items by 2千円, and 13 of 20 relationships fail exact equality although the page is correct. A fixed tolerance is either too loose (it hides real errors) or too tight (it raises false alarms). Instead, each printed value is treated as the interval of true values it could represent, following the XBRL Calculations 1.1 specification, which checks consistency by interval overlap and supports both truncation and round-to-nearest [10]:

> truncation: $v \geq 0 \to [v, v{+}1)$; $v < 0$ (△) $\to (v{-}1, v]$; △$0 \to (-1, 0)$; $- \to \{0\}$
> rounding: $v \to [v{-}½, v{+}½]$
> a relationship holds if and only if the interval of the reported total overlaps the interval sum of its terms.

The tolerance then emerges from the number of terms and the rounding mode rather than from a chosen constant. The rounding mode can also be inferred. Under truncation, the reported total minus the sum of its printed parts must lie between 0 and $n{-}1$ for sums of positive terms, and between −1 and 0 for a net amount computed as gross minus accumulated depreciation. Every residual on the sample page has exactly that sign pattern, which identifies truncation (千円未満切捨て), whose Calculations 1.1 intervals are half-open.

6.2 Where constraints come from

1. **The table's own tree.** A section's closing total equals the sum of its members' contributions; a net row equals its gross row plus its contra rows. No prior knowledge is needed, which is why this works on day one and on unseen companies.
2. **Taxonomy calculation links.** Once rows are mapped to concepts, the taxonomy's addition and subtraction relationships [14] supply constraints for statements that are not simple trees, such as income-statement step-downs.
3. **Statement templates.** The statement of changes in equity is a matrix: each column's closing balance equals opening balance plus movements, and each row's components sum across columns. Two-dimensional constraints of this kind pin down almost every cell.
4. **Cross-statement ties.** Total assets equal liabilities plus net assets; closing equity in the equity statement matches the balance sheet; profit ties between the income and equity statements. Cash in the cash-flow statement is compared with balance-sheet cash as a soft check, because the definitions differ.
5. **Cross-document checks.** The prior-year column against last year's filing (soft, because of restatements); the earnings-release summary page, usually in 百万円, against the statements, which catches scale errors; the

earnings release against the annual report; and for corrections, the requirement that the 訂正前 to 訂正後 difference touches only underlined cells.

6. **XBRL.** Where the filing's XBRL is available, it is the strongest independent source (Section 7).

6.3 Which check catches which error

No single check covers every error type, so the design assigns each error type at least one strong detector. Measured figures come from the mutation test in Section 8; the other entries are the expected coverage, to be measured on the evaluation set.

Error type	Grounding	Structure	Arithmetic	Path agreement	Taxonomy and cross-document	XBRL
Wrong digit, 10 units or more	–	–	● 100%	●	◐	●
Wrong units digit	–	–	◐ 47–71%	●	◐	●
Lost or spurious △	–	–	● 100%	◐	◐	●
Values swapped between adjacent rows	–	–	◐ 45%	●	◐	●
Period columns swapped	–	◐	● 100%	◐	●	●
Hallucinated value or row	●	◐	●	●	–	●
Dropped row or value	–	◐	● unless nil	●	–	●
Wrong scale (千円 read as 百万円)	–	◐	– scale-invariant	◐	●	●
Wrong label characters	–	–	–	●	◐	●
Wrong parent in hierarchy	–	◐	●	◐	●	◐
Missed page continuation	–	● unclosed group	◐	–	◐	◐
Wrong scope (連結 read as 個別)	–	◐	–	–	●	●

● strong detector ◐ partial – none. Percentages are the measured detection rates of the prototype's arithmetic checks on the sample page.

Two rows deserve emphasis. Scale errors are invisible to every sum check, because multiplying all values by 1,000 preserves every identity; only cross-document comparison or XBRL catches them. Label errors are invisible to arithmetic altogether. Both are reasons the design does not rely on arithmetic alone.

6.4 Arbitration and page states

State	Condition
Verified	Every value grounded; paths agree, or a single path passed its audit; all hard constraints pass; every cell is covered by at least one constraint or confirmed by an independent source.
Accepted with warnings	Values verified, but a soft check failed with a plausible explanation (for example a restated prior-year column), or a concept mapping is uncertain. The flag travels with the facts.
Needs review	An unresolved disagreement between paths, a failed hard constraint, or a cell that no check covers and no second source confirms.
Failed	The page cannot be parsed, for example an unreadable scan. Reported with the reason.

Constraint-guided repair. When the paths disagree on exactly one cell and exactly one candidate reading satisfies every constraint, that reading is accepted and marked as repaired. A repair may choose between values that a path actually read; it never computes a value from the constraint, because that would replace evidence with inference.

7. Evaluation and monitoring

7.1 Building ground truth

XBRL-aligned gold, automatically. Most recent statement PDFs have a machine-readable twin. XBRL filing on TDnet began with the new TDnet in 2008 [12], and earnings-release financial statements are now filed as inline XBRL that extends the EDINET taxonomy [11]. EDINET's API v2 serves annual-report XBRL and a converted tabular form (type 5, UTF-16 tab-separated) [13]. Aligning each extracted fact with an XBRL fact by period, value and concept or label yields gold labels for thousands of pages at almost no cost. Disagreements are adjudicated by hand, because XBRL can itself differ from the PDF (for example when a correction is filed separately); each disagreement is a finding, not an assumption.

A hand-verified stratified set covers what XBRL cannot: pre-XBRL and scanned filings, layout fidelity, hierarchy and metadata. **A synthetic stress set** renders statements from known XBRL facts into controlled variations (fonts, spaced labels, account form, rotation, scan noise and skew), so gold is known by construction and each complexity in Section 3 can be tested in isolation. Existing open resources help: EDINET-Bench ships tooling to download and parse EDINET filings [16], and the NTCIR-18 U4 task provides HTML tables from securities reports, where layout features proved the second most important input after text [17].

7.2 Stratification

Results are reported per stratum: era, producer family, filing type, statement type, accounting standard, industry, digital or scanned, and single or multi-page. Producer family matters more than it first appears. Japanese disclosure documents are largely prepared with a small number of vendor systems; one vendor group describes its system as holding more than half the market, and another operates its own platform [23]. Layouts therefore cluster into families, which makes family-level regression suites effective. It also means a vendor template change can shift a whole family at once, so drift is monitored per family.

7.3 Monitoring in production

Production has no gold labels, so monitoring uses proxies, tracked per family and per week: constraint failure rate, path disagreement rate, review rate and repair rate. A jump in any of them for one family signals a layout change before users notice. The only unbiased estimate of residual error is a random audit: a fixed share of verified pages is reviewed by people every week, and the rule of three sets the sample size needed for the precision we want to claim.

8. Evidence from the prototype

The prototype implements Path A, the tree-derived constraints, the interval-arithmetic validator and an evaluation harness, in about 1,650 lines of Python (in the accompanying repository). It was developed on the single sample page, so its results show that the approach works and how strong the checks are; they do not show generalisation, which is what Section 7 is for.

Measure on the sample page	Result
Metadata: statement title, consolidated scope, correction status, unit 千円, two periods with dates	all correct
Row structure (label and hierarchy path), F1	1.00 (43 / 43)
Facts (path, period end, typed value), F1	1.00 (74 / 74)
Correction underlines detected	2 / 2
Constraints derived from the tree, passing	20 / 20
Cells covered by at least one constraint	100%
Exact-equality checks failing on the same correct page	13 / 20
pdfplumber default: body rows reproduced, but no periods, unit or hierarchy	43 / 43
pdfplumber text strategy: body rows reproduced; labels split mid-word (現 | 金及び預金)	39 / 43

How strong is "all checks pass"? To answer that, the harness injects one realistic error at a time into the correct table (every digit substitution at every position, OCR-confusable digits, dropped digits, lost △ signs, swapped period columns and swapped adjacent rows, 4,199 mutants in all) and records whether any constraint fails.

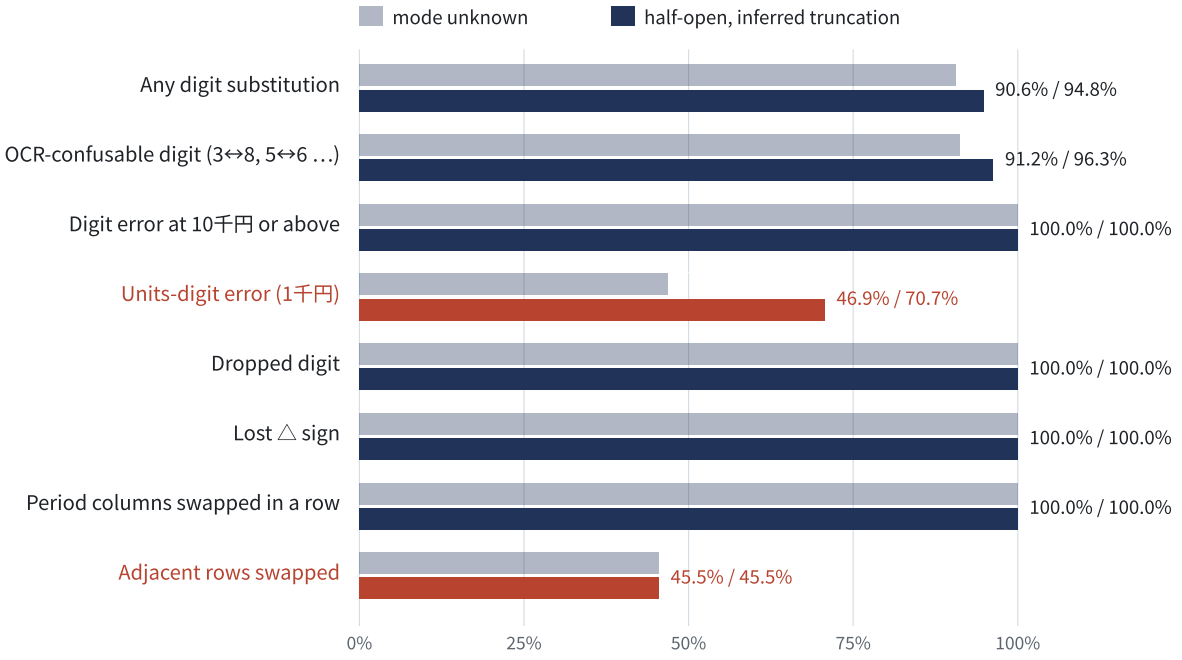


Figure 3. Share of injected errors detected by the arithmetic checks alone, when the rounding mode is unknown (hull of the truncation and rounding intervals) and with the half-open truncation intervals of XBRL Calculations 1.1 once truncation is inferred. Every digit error at 10千円 or above is caught. Units-digit errors and swaps between rows that feed the same total are the blind spots.

The result is reassuring and sobering in equal measure. Any digit error at 10千円 or above, any dropped digit, any lost △ and any swap of period columns is caught every time. But a units-digit error is caught only 47% of the time with the mode unknown and 71% with truncation inferred, because it can hide inside the rounding interval. And when two values that feed the same total are swapped, the total is unchanged, so arithmetic catches only 45% of adjacent-row swaps. These two blind spots are exactly the cells where the second extraction path and XBRL earn their place, as the coverage matrix in Section 6.3 shows.

The prototype also surfaced the traps of Figure 1 without being told about them: background-coloured rule segments that a colour-blind parser would treat as cell borders, short correction underlines that were initially mistaken for summation rules, and tiny junction slivers in the rule artwork. Each led to a specific rule (filter by fill

colour; require a column rule to span 90% of the column; require an underline to match glyph width), which is the kind of case the stratified regression suite is meant to pin down.

9. Risks, trade-offs and roadmap

Risks

- **Gold coverage gaps.** Older and scanned filings lack XBRL, which is where errors are most likely. The hand-verified and synthetic sets exist to cover them, and their size is set by the confidence we need per stratum.
- **Model licensing and cost.** The strongest Japanese-specific engine found is non-commercial [9]; commercially licensed options must be evaluated on our strata. Running the vision path on every page is expensive, so on audited digital pages it runs selectively: on value-column digits always, and fully only when verification is incomplete.
- **Taxonomy and format drift.** The EDINET taxonomy is revised regularly and disclosure rules change (the 2024 quarterly-reporting reform changed TDnet's filing specifications [11]). Concept mappings and templates are versioned by taxonomy year.
- **Correlated errors.** Both paths can fail on the same faint △ or the same degraded scan. Arithmetic and cross-document checks are the backstop, and the random audit measures what remains.
- **Review capacity.** Thresholds are tuned against reviewer throughput, so the review queue is sized to the team that works it, and coverage targets are reported together with error rates.

Trade-offs chosen

Geometry rules are brittle on unseen layouts but exact and explainable where they apply; vision models generalise but can misread. Pairing them under one verifier keeps the precision of the first and the reach of the second. Strict intervals catch more units-digit errors but depend on correctly inferring the rounding mode, so strict mode is applied only when the residual pattern confirms it. Abstaining costs automation rate but protects the error rate, which is the number that matters for financial data.

Roadmap

Phase	Scope	Exit criterion
1	Digital J-GAAP earnings releases: balance sheet, income statement, cash flows. Path A, tree constraints, XBRL-aligned evaluation harness.	Fact error at or below 0.1% among verified facts on the digital stratum, with confidence intervals
2	Concept mapping with taxonomy calculation links; cross-page assembly; income and cash-flow templates; review interface.	Multi-page statements assembled; automation rate measured per family
3	Path B for scans and untrusted text layers; text-layer audit; agreement and repair.	Scanned stratum meets the same error target, at lower automation
4	IFRS and US GAAP; equity-statement matrices; industry formats; account-form pages; notes tables; production drift monitoring.	All strata reported; weekly audit in place

Appendix A. Output schema

Each fact is self-describing and carries its provenance and verification record. A rendered table is derived from facts, never the other way round.

```
{
  "fact_id": "sha1:…",
  "statement": {"type": "balance_sheet", "scope": "consolidated", "standard": "jp_gaap",
                "filing": "tanshin", "correction": "after"},
  "row": {"label_raw": "現金及び預金", "label_norm": "現金及び預金",
          "path": ["資産の部", "流動資産", "現金及び預金"], "role": "item",
          "concept": "jppfs_cor:CashAndDeposits", "concept_confidence": 0.99},
  "period": {"type": "instant", "end": "2026-03-31", "label": "当連結会計年度"},
  "value": {"reported": 608281, "unit": "JPY", "scale": 1000, "jpy": 608281000,
            "interval_jpy": [608281000, 608282000), "sign": "+", "state": "amount",
            "rounding": "truncate"},
  "flags": {"corrected": false, "footnote_refs": []},
  "provenance": {"doc_sha256": "…", "page": 4, "bbox": [490.9, 172.2, 522.6, 181.2],
                 "extractor": "pathA-geometry@0.3", "agreement": "pathB-vision@1.6"},
  "verification": {"state": "verified", "constraints_passed": ["sum:流動資産合計"],
                   "xbrl_match": true}
}
```

The interval uses mathematical notation for readability; it is serialised as bounds with open or closed flags. Nil (—) and blank (not presented) are distinct values of `state`.

Appendix B. Glossary

Term	Meaning
決算短信 (tanshin)	Earnings release filed with the exchange through TDnet
有価証券報告書	Annual securities report filed with the FSA through EDINET
連結 / 個別	Consolidated / non-consolidated
貸借対照表, 財政状態計算書	Balance sheet (J-GAAP), statement of financial position (IFRS)
勘定式 / 報告式	Account form (side-by-side) / report form (vertical) layout
前・当連結会計年度	Prior and current consolidated fiscal year (column headers)
単位：千円	Unit line: thousands of yen
△ / ▲	Negative amount markers
ー	Nil: the line exists with no balance
合計, 小計, 純額	Total, subtotal, net amount
減価償却累計額	Accumulated depreciation (a contra row)
千円未満切捨て	Amounts truncated below one thousand yen
訂正前 / 訂正後	Before / after correction
均等割付	Distributed spacing of a label across a fixed width
和暦	Japanese era dates (令和, 平成)

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